📈 Phase 3: Sales Trends & Logistics Performance (Summary)

In the final phase of this project, I developed specialized analytical tables (**Data Marts**) to provide a structured overview of business growth and operational efficiency. This stage focuses on delivering high-quality, aggregated data ready for further business analysis.

1. Technical Implementation: Building Data Marts

- **Strategic Aggregation:** I transformed raw transactional data into high-level summary tables. This process reduces data complexity and ensures that key business metrics are pre-calculated and consistent.
- **Sales Trends Analysis:** I developed a dedicated table to track monthly revenue, order volume, and unique customer growth. This enables the identification of seasonal performance patterns, such as the peak recorded in November 2017.
- **Logistics Performance Metrics:** I implemented a tracking system for delivery health by calculating the precise difference between estimated and actual delivery dates.
- **Data Quality Assurance:** The SQL script uses advanced filtering to include only completed deliveries with valid timestamps, ensuring the reliability of the final analytical output.

2. Key Business Insights

- **Growth Dynamics:** The analysis reveals a strong upward trend throughout 2017, with revenue peaking at over **$1.15 million** in November. This highlights the massive impact of seasonal sales events on the platform's turnover.
- **Regional Disparities:** São Paulo (SP) stands out as the most efficient hub, managing over **42,000 orders** with high operational stability.
- **Delivery Efficiency:** On average, packages are delivered 8 to 21 days before the estimated date. However, the analysis also identifies specific regions where the late order rate (pct_late) exceeds **20%**, pointing to local logistical challenges.

3. Strategic Recommendations

- **Supply Chain Optimization:** Use the generated logistics table to target improvements in states with high delay rates, such as Alagoas (AL).
- **Market Blueprint:** Leverage the data from the São Paulo region to establish performance benchmarks for other regional markets.